THE ROTATED TRIANGLE OF AN ADMISSIBLE SHORT EXACT SEQUENCE IS ISOMORPHIC TO THE CONE OF ITS CONNECTING MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-rotate-triangle (source lines 978-995); the proof environment begins two lines below it and is its own proof, closing at source line 1013.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the cone construction and the associated triangle are the author's own surrounding development in the same chapter.

Editorial note. The proof displays one block matrix and one displayed pair of triangles; both are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the rotated triangle of an admissible short exact sequence is isomorphic to the cone of its connecting morphism.

Allowed standard prerequisites (editorial).

- Cone of a morphism of dg modules. The cone C(f) of a morphism f : X -> Y is the dg module Y direct sum X[1] with differential given by the block matrix of d_Y and the shifted d_X against f.. It gives the differential matrix of C(delta[-1]) that is compared with d_B (source lines 1006-1012).
- Associated triangle of an admissible short exact sequence. An admissible short exact sequence 0 -> K -> L -> M -> 0 of dg modules determines a morphism delta : M -> K[1] and a triangle (K, L, M, alpha, beta, delta).. It supplies the connecting morphism delta whose shift enters the matrix of d_B (source lines 1003-1006).
- Splitting of an admissible sequence. An admissible short exact sequence of dg modules splits as graded modules, so the middle term can be written as the direct sum of the outer terms.. The proof chooses splittings to write L = K direct sum M and to identify alpha and beta with the inclusion and projection (source lines 998-1000).

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let $\alpha : K \to L$ and $\beta : L \to M$ define an admissible short exact sequence*

$$0 \to K \to L \to M \to 0$$

of differential graded A-modules. Let $(K, L, M, \alpha, \beta, \delta)$ be the associated triangle. Then the triangles

$$(M[-1], K, L, \delta[-1], \alpha, \beta) \quad \text{and} \quad (M[-1], K, C(\delta[-1]), \delta[-1], i, p)$$

are isomorphic.

Proof. Using a choice of splittings we write $L = K \oplus M$ and we identify α and β with the natural inclusion and projection maps. By construction of δ we have

$$d_B = \begin{pmatrix} d_K & \delta \\ 0 & d_M \end{pmatrix}$$

On the other hand the cone of $\delta[-1] : M[-1] \to K$ is given as $C(\delta[-1]) = K \oplus M$ with differential identical with the matrix above! Whence the lemma. $\square$